KrishiBot: A YOLOv8n-Powered Vision System for Autonomous Spraying in Weed Management

***Abstract*— Weed proliferation presents a major challenge to sustainable agriculture, particularly in developing countries where manual weeding is labor-intensive and automation solutions are often financially or technologically inaccessible. In this paper, we propose a real-time weed segmentation and classification framework designed for deployment in resource-constrained environments. Our system is built on YOLOv8 architecture, trained on a real-time dataset of locally captured images enhanced through advanced preprocessing and augmentation techniques. Canny edge detection, CLAHE contrast enhancement, and synthetic data generation using manual segmentation have been incorporated to enrich training data and improve edge awareness. Additionally, the segmentation model was deployed on a customized autonomous spraying bot that communicates via a master-slave protocol between a Raspberry Pi and an Arduino. The proposed solution demonstrates robust generalization across varied backgrounds and lighting conditions, proving to be both lightweight and effective for real-time application in rural agriculture.**

Keywords—YOLOv8, Weed Segmentation, Master-slave Protocol, Lightweight, Edge devices, Real-time Dataset

I. INTRODUCTION

Weeds significantly hinder agricultural productivity by competing with crops for vital resources such as water, sunlight, and nutrients. Weeds typically compete with crops for essential resources during the initial one-third of the crop's life cycle. Therefore, timely removal during this critical growth phase is essential to minimize yield loss and ensure optimal crop development. Traditional weed management practices often rely on extensive manual labor, which is time-consuming, inefficient, and economically burdensome for rural communities. With the rise of deep learning, computer vision-based weed detection systems have shown great promise; however, most existing solutions are computationally intensive and unsuitable for deployment on edge devices commonly used in low-resource environments.

Segmentation-based approaches offer finer granularity compared to object detection, enabling pixel-level weed identification even in cluttered or low-resolution scenarios. In this study, we present a lightweight, real-time instance segmentation model based on a YOLOv8 architecture, optimized for both accuracy and efficiency. Our dataset, collected from agricultural fields in Bangladesh, includes diverse real-world weed samples under different lighting and environmental conditions. To further enhance model performance, we employ Canny edge detection to amplify boundary features, integrate histogram equalization via CLAHE, and transform images into alternative color spaces to highlight subtle contrasts. To address dataset limitations, synthetic images were generated by manually extracting weed masks and overlaying them onto a variety of local background images, effectively increasing diversity and robustness. This software stack is paired with a hardware prototype: an autonomous spraying robot constructed using Raspberry Pi and Arduino modules in a master-slave configuration. For deployment on the Raspberry Pi, the model was converted to the NCNN framework, which is optimized for ARM-based architectures and performs inference using low-level C++ implementations. The entire system has been tested in real-time on field-like testbeds, demonstrating high responsiveness and inference accuracy under operational constraints.

II. LITERATURE REVIEW

The application of deep learning in agriculture has gained momentum in recent years, particularly in tasks like disease detection, crop classification, and weed identification. However, most existing research either depends on high-performance GPU resources or utilizes generic datasets that fail to capture the complexity of local field environments.

Rai et al. [1] addressed these challenges by integrating deep learning with edge computing to detect agricultural weeds in real time. Their work emphasized the need for lightweight architectures capable of running on portable devices. Similarly, Chamara et al. [2] introduced AICropCAM, a modular platform combining classification, segmentation, and object counting, capable of running on low-power hardware with promising results in crop monitoring.

Although several studies, such as Dai [3], explored real-time object detection using quantized models like TensorFlowLite, and the shift from object detection to segmentation remains underexplored in weed management. Segmentation offers pixel-level precision necessary for tasks such as targeted spraying, which is crucial in reducing chemical usage. Despite the effectiveness of architectures like U-Net and Mask R-CNN, their high computational demands often make them unsuitable for field deployment.

Sethia et al. (2020) [4] developed an automated weed removal robot that integrates computer vision and robotics to address the challenges of manual weeding and indiscriminate herbicide use. Their system employs a Raspberry Pi-based platform with a downward-facing camera to capture images, which are then processed using Mask R-CNN for semantic segmentation and VGG16-based CNN for accurate weed-crop classification, achieving a 99.5% accuracy rate. The robot features a delta arm with a high-speed rotating blade to physically remove detected weeds, minimizing chemical usage and environmental impact. The hardware is designed for efficient field traversal and is powered by a hybrid battery-solar system.

Our work builds on these foundations by using a modified YOLOv8 model tailored for segmentation and optimized

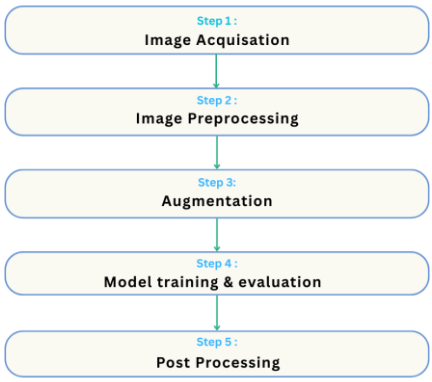


Fig. 1. Training the segmentation model.

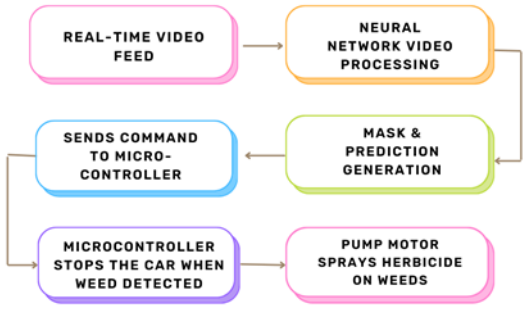


Fig. 2. Flow chart of our KrishiBot.

through model compression techniques for edge inference. Additionally, few existing studies incorporate preprocessing techniques such as Canny edge enhancement or synthetic image generation—both of which are central to our pipeline. By capturing real-time data, applying context-aware augmentation, and integrating the model into a real-time spraying bot, this study provides a comprehensive and deployable solution for intelligent weed management in resource-constrained settings.

III. Proposed Methodology

Our workflow can be divided into 2 major segments, namely Hardware and Software. The workflow for the Software part & hardware part is illustrated in Fig. 1 and Fig. 2 respectively.

The following subsections provide more information about the suggested methodology.

A. Data Collection

The effectiveness of a segmentation model in real-time agricultural settings heavily depends on the quality and relevance of the training dataset. For this work, images were captured from real weed-infested fields during a visit to Bangladesh Agricultural University (BAU), Mymensingh. Collaboration with the Department of Agronomy facilitated access to the Weed Museum and nearby jute fields. The data captured includes high- and low-resolution images under varying lighting and environmental conditions to simulate real-field challenges such as occlusion, blur, and irregular lighting. Video recordings were also made and frames extracted to diversify the image pool and include temporal variations commonly observed in motion capture. A total of around 600 base images were collected for our dataset. Sample images are shown in Fig. 3.



Fig. 3. Sample data.

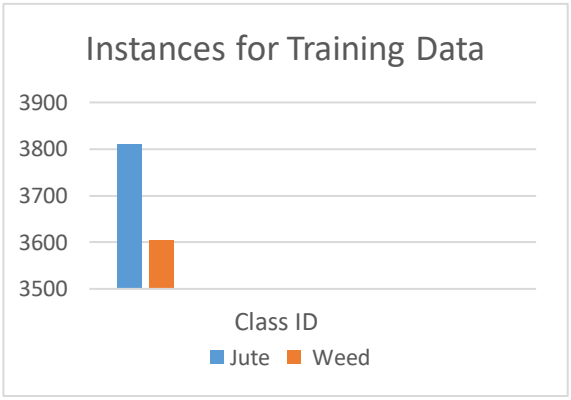


Fig 4. Visualization of class instances.

B. Data Annotation

Each collected image was meticulously annotated using the Roboflow platform. The annotation process involved generating segmentation masks for various weed species, following a class-wise labeling convention compatible with YOLO segmentation format. Roboflow's vision APIs were employed to streamline annotation, and the final dataset was exported in a format directly ingestible by the training pipeline. Special attention was given to differentiating overlapping and co-occurring weed species to ensure accurate multi-class segmentation. All of the data in the dataset were categorized into 2 classes: Jute and Weed.

C. Data Preprocessing

To improve model accuracy and boundary awareness, a hybrid preprocessing strategy was applied. Canny edge detection was first used to extract strong structural outlines from each image. These edge maps were merged into the red channel of the original RGB image to enhance spatial feature representation during training. Additionally, Contrast Limited Adaptive Histogram Equalization (CLAHE) was applied to improve local contrast in varying lighting conditions. Images were also transformed into the LHS color space to capture perceptual features more robustly than the standard RGB domain.

Due to limitations in dataset size, multiple augmentation strategies were applied including Gaussian noise addition, random cropping, shearing, rotation, and color saturation adjustments. Furthermore, synthetic images were created by extracting foreground weed masks and overlaying them on new background scenes collected from the surrounding rural environments. This manual signal processing-based synthesis enriched the dataset's diversity, making the model more robust to spatial and textural variations. The final version of the dataset contained about 1802 images that were split into train, test and validation sets according to the ratio 0.7:0.2:0.1. The number of instances of the 2 classes in our dataset can be visualized according to Fig. 4.

D. Model Selection and Training

Two segmentation models were evaluated in this work: **YOLOv8/YOLOv11-based segmentation architectures** and **Mask R-CNN**.

YOLOv8 & YOLOv11 Segmentation: The YOLO (You Only Look Once) family of models is renowned for its speed and accuracy. YOLOv8 and its successor YOLOv11 include segmentation heads capable of producing pixel-level masks. Their architecture integrates a CSPDarknet backbone, PANet-based feature aggregation, and optimized decoders for segmentation tasks. These models are well-suited for edge deployment due to their low-latency inference and reduced memory footprint. In our experiments, the YOLOv8s, YOLOv8m, YOLOv11s, YOLOv11m, and YOLOv11l variants were trained using annotated data. Training was performed on both Kaggle GPUs and local systems equipped with RTX 3050/3060, with hyperparameters: image size = 640, batch size = 16, epochs = 100. Among the variants, **YOLOv11m** showed the best trade-off between accuracy and inference speed. It delivered high mean Average Precision (mAP@0.5) and strong generalization to new field conditions while maintaining compatibility with embedded deployment.

E. Hardware Implementation

To realize a complete edge-enabled smart weed management system, we integrated our trained model into an **AI-based autonomous herbicide spraying bot (KrishiBot)**. The bot was built using a **Raspberry Pi 4 (8GB)** as the master controller and an **Arduino Mega 2560** as the slave unit. The Raspberry Pi processed the image input using a **Pi Camera Module V2 (8MP)** and ran the segmentation model in real time.

The Arduino controlled the mechanical operations - receiving instructions from the Raspberry Pi through UART communication. Motion was driven using **37GB-550 DC motors (100 RPM)**, actuated via **L293D motor drivers**. The bot was equipped with a **12V 10L/min water pump** to simulate herbicide spraying when a weed was detected. Power was supplied by dual 12V 1.3A batteries, and off-road wheels were used for movement across uneven field surfaces.

This modular architecture enabled real-time detection and spraying, tested on a field-like testbed. The hardware setup demonstrated successful weed segmentation followed by localized actuation, showcasing the feasibility of integrating lightweight AI models into precision farming equipment for rural deployment. The final assembled bot and the test bed is illustrated in Fig. 5 and Fig. 6, respectively.

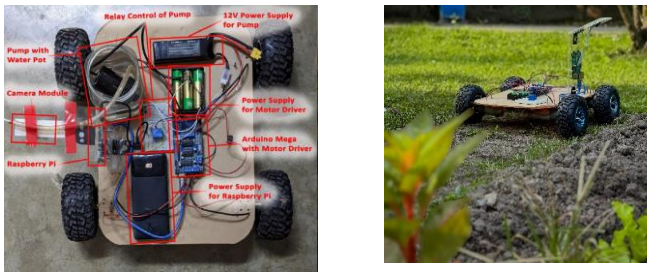


Fig 5. KrishiBot.



Fig 6. Test Bed.

IV. Proposed Model

YOLO (You Only Look Once) is a high-speed, real-time object detection framework that has evolved to support semantic segmentation in its latest versions. YOLOv8 introduces an encoder-decoder-based architecture that simultaneously performs object classification, bounding box regression, and instance mask prediction. It is widely appreciated for its balance of performance and efficiency, making it a strong candidate for real-time, edge-friendly segmentation tasks.

A. Architectural Overview

The base YOLOv8 segmentation architecture used in this work consists of three key components:

Backbone: A CSPDarknet-based convolutional structure that extracts multi-scale features from input images while maintaining a compact representation through Cross Stage Partial (CSP) connections.

Neck: A PANet-inspired Path Aggregation Network that strengthens feature propagation across layers, particularly from low-level spatial features to high-level semantic representations.

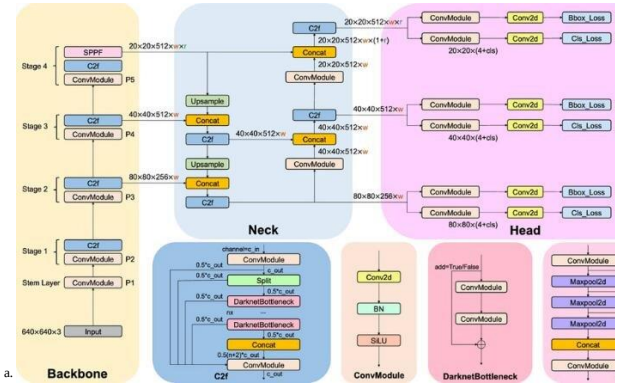
Head: A multi-output detection and segmentation head that performs three tasks: objectness scoring, class prediction, and binary mask generation for each detected object.

B. Model Variants and Customization

Multiple variants of YOLOv8 were explored, including YOLOv8s and YOLOv8m, along with upgraded YOLOv11-based backbones. Among these, we selected and further refined **YOLOv11m** for its optimal trade-off between network depth and computational complexity. The customizations applied to this variant included:

contrast features (Canny + CLAHE + LHS), enabling enhanced feature representation.

Synthetic Data Support: The model was exposed to a hybrid dataset containing both original and synthetically composed weed images, increasing its resilience to domain shift and environmental noise.



[5]

Fig 7. YOLO V8 architecture.

These customizations were implemented so that the model can learn contextual patterns specific to weeds found in Bangladeshi agricultural fields.

Input Preprocessing-aware Adaptation: The model was trained on preprocessed images with embedded edge and Integration Readiness

The final YOLOv8 model was designed with edge deployment in mind. We used the nano version of the model and also exported it from Pytorch format to NCNN format, which is more optimized for ARM processors. Lightweight convolution layers were favored, and channel-wise operations were limited to optimize inference speed. This proposed model stands at the core of the weed segmentation pipeline and forms the intelligence behind the hardware system described earlier. The next section details the results and evaluation metrics that demonstrate its performance.

V. Result and Analysis

A. Evaluation Metrics

To evaluate the segmentation performance of the proposed model, several standard metrics were used to assess different aspects of accuracy, overlap, and classification quality:

Intersection over Union (IoU): Measures the overlap between predicted masks and ground truth.

$$\text{IoU} = \frac{\text{TP}}{(\text{TP}+\text{FP}+\text{FN})} \quad (1)$$

F1 Score: Harmonic mean of precision and recall, providing a balanced assessment of false positives and false negatives.

$$\text{F1} = \frac{2*\text{Precision}*\text{Recall}}{\text{Precision}+\text{Recall}} \quad (2)$$

Recall: Indicates the model's sensitivity—its ability to correctly identify true weed regions.

$$\text{Recall} = \frac{\text{TP}}{\text{TP}+\text{FN}} \quad (3)$$

Mean Average Precision (mAP@0.5): The mean precision score across all classes at a fixed IoU threshold 0.5.

These metrics are calculated from the confusion matrix which is given in Fig-9.

TABLE I . SEGMENTATION PERFORMANCE

Model	F1 Score	Recall	Precision	mAP@0.5
YOLOv8n	0.31	0.6	0.3	22%
YOLOv8m	0.31	0.58	0.34	21.2%
YOLOv11s	0.30	0.62	0.32	23.8%
YOLOv11m	0.29	0.57	0.35	20.1%
YOLOv11l	0.34	0.6	0.38	26%

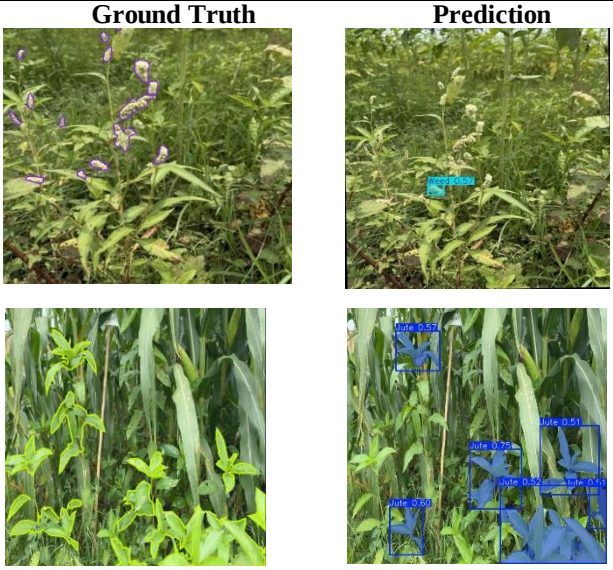


Fig 8. Sample prediction.

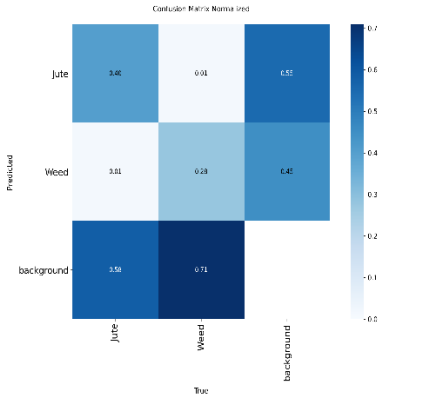


Fig 9. Confusion matrix.

B. Comparison Analysis

Several model variants from the YOLO family were evaluated for benchmarking purposes. Table-I summarizes the relative performance.

We can see that, the model struggled to properly classify probably due high complexity of dataset. Transformer based models or Vision Language Model(VLM) might have been more appropriate for this task, but none of them were deployable on a Raspberry Pi.

Although **YOLOv11l** achieved marginally higher accuracy, their inference time and memory usage were significantly higher, making them less suitable for real-time field deployment on edge hardware. In contrast, **YOLOv11s** offered an optimal balance between accuracy, speed, and hardware compatibility.

Visual comparisons of predicted segmentation masks confirmed YOLOv11s's effectiveness in preserving fine

weed boundaries and maintaining class consistency across cluttered backgrounds. The use of Canny edge-enhanced training images and synthetic dataset augmentation contributed significantly to the model's superior generalization and object boundary alignment.

However, to ensure real-time inference on resource-constrained hardware, we selected the **YOLOv8n** model, as it was the only variant sufficiently lightweight to achieve acceptable frames per second (FPS) on a Raspberry Pi. Representative prediction results generated by the **YOLOv8n** model are illustrated in Fig.8.

The full demonstration video on custom test bed is given here:
https://youtube.com/shorts/WFLUrrWUYRI?feature=share

VI. Conclusion

This work presents a comprehensive real-time weed segmentation and spraying solution tailored for resource-constrained agricultural environments. By merging two complementary projects—one focusing on pixel-level weed segmentation using a customized YOLOv8 architecture and the other on a hardware-controlled herbicide spraying bot—we offer a unified system capable of identifying and responding to weed presence with minimal human intervention.

The segmentation model, trained on a locally curated and synthetically expanded dataset, integrates preprocessing techniques such as edge enhancement and contrast normalization to improve feature awareness in field conditions.

Complementing the software pipeline, the hardware system incorporates a Raspberry Pi–based control unit communicating with an Arduino Mega to drive motion and activate spraying mechanisms. The entire setup was tested in realistic testbed environments, validating its feasibility for precision agriculture applications.

By focusing on segmentation rather than object detection, the system achieves finer spatial resolution and more accurate weed localization, enabling targeted intervention. This approach not only enhances herbicide efficiency but also contributes to sustainable agricultural practices by reducing overuse of chemicals.

In future iterations, we aim to expand the dataset with seasonal variability, integrate adaptive spraying mechanisms, and deploy the system in larger field trials for longitudinal evaluation.

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